

Quiz1_solutions

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Linear Regression with the Ames Housing Data – Regression Task

The Ames_Housing_Data.csv file contains data on various aspects of houses sold. Our goal is to predict the SalePrice using all the other variables.

```
if (!requireNamespace("tidyverse")) install.packages('tidyverse')

## Loading required namespace: tidyverse

if (!requireNamespace("caret")) install.packages('caret')

## Loading required namespace: caret

if (!requireNamespace("leaps")) install.packages('leaps')

## Loading required namespace: leaps

if (!requireNamespace("bestglm")) install.packages('bestglm')

## Loading required namespace: bestglm

if (!requireNamespace("MASS")) install.packages('MASS')
library(tidyverse)

## Warning: package 'ggplot2' was built under R version 4.4.3

## Warning: package 'tibble' was built under R version 4.4.1

## Warning: package 'tidyr' was built under R version 4.4.1

## Warning: package 'purrr' was built under R version 4.4.1

## Warning: package 'lubridate' was built under R version 4.4.1

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    4.0.1      v tibble     3.3.0
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag() masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.4.1
```

```
## Loading required package: lattice
```

```
## Warning: package 'lattice' was built under R version 4.4.1
```

```
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
## lift
```

```
library(leaps)
library(bestglm)
library(MASS)
```

```
## Warning: package 'MASS' was built under R version 4.4.1
```

```
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
## select
```

```
data = read.csv('Ames_Housing_Data.csv')
str(data)
```

```
## 'data.frame': 1460 obs. of 20 variables:
## $ Id : int 1 2 3 4 5 6 7 8 9 10 ...
## $ LotArea : int 8450 9600 11250 9550 14260 14115 10084 10382 6120 7420 ...
## $ OverallQual : int 7 6 7 7 8 5 8 7 7 5 ...
## $ OverallCond : int 5 8 5 5 5 5 5 6 5 6 ...
## $ YearBuilt : int 2003 1976 2001 1915 2000 1993 2004 1973 1931 1939 ...
## $ YearRemodAdd : int 2003 1976 2002 1970 2000 1995 2005 1973 1950 1950 ...
## $ CentralAir : chr "Y" "Y" "Y" "Y" ...
## $ X1stFlrSF : int 856 1262 920 961 1145 796 1694 1107 1022 1077 ...
## $ X2ndFlrSF : int 854 0 866 756 1053 566 0 983 752 0 ...
## $ GrLivArea : int 1710 1262 1786 1717 2198 1362 1694 2090 1774 1077 ...
## $ FullBath : int 2 2 2 1 2 1 2 2 2 1 ...
## $ HalfBath : int 1 0 1 0 1 1 0 1 0 0 ...
## $ BedroomAbvGr : int 3 3 3 3 4 1 3 3 2 2 ...
## $ KitchenAbvGr : int 1 1 1 1 1 1 1 1 2 2 ...
```

```
## $ TotRmsAbvGrd: int 8 6 6 7 9 5 7 7 8 5 ...
## $ Fireplaces : int 0 1 1 1 1 0 1 2 2 2 ...
## $ GarageCars : int 2 2 2 3 3 2 2 2 2 1 ...
## $ GarageArea : int 548 460 608 642 836 480 636 484 468 205 ...
## $ YrSold : int 2008 2007 2008 2006 2008 2009 2007 2009 2008 2008 ...
## $ SalePrice : int 208500 181500 223500 140000 250000 143000 307000 200000 129900 118000 ...
```

```
table(data$CentralAir)
```

```
##
##      N      Y
## 95 1365
```

```
data$CentralAir = ifelse(data$CentralAir == 'Y', 1, 0)
```

```
# 1) 15 pts
```

```
set.seed(123)
```

```
training.samples = data$SalePrice %>%
```

```
  createDataPartition(p = 0.75, list = FALSE)
```

```
train.data = data[training.samples, ]
```

```
test.data = data[-training.samples, ]
```

```
# 2) Stepwise BIC, 30 pts
```

```
full = lm(SalePrice ~., data = train.data)
```

```
null = lm(SalePrice ~ 1, data = train.data)
```

```
n = nrow(train.data)
```

```
best.model1 = stats::step(null, scope = list(lower = null, upper = full), direction = 'both', k = log(n))
```

```
## Start: AIC=24805.6
```

```
## SalePrice ~ 1
```

```
##
```

	Df	Sum of Sq	RSS	AIC
## + OverallQual	1	4.5678e+12	2.6399e+12	23711
## + GrLivArea	1	3.9340e+12	3.2737e+12	23947
## + GarageCars	1	2.9822e+12	4.2255e+12	24227
## + GarageArea	1	2.8966e+12	4.3111e+12	24249
## + X1stFlrSF	1	2.8508e+12	4.3569e+12	24260
## + FullBath	1	2.3153e+12	4.8924e+12	24388
## + TotRmsAbvGrd	1	2.2046e+12	5.0031e+12	24412
## + YearBuilt	1	1.9200e+12	5.2877e+12	24473
## + YearRemodAdd	1	1.8773e+12	5.3304e+12	24482
## + Fireplaces	1	1.5633e+12	5.6444e+12	24544
## + X2ndFlrSF	1	7.3966e+11	6.4680e+12	24694
## + LotArea	1	5.9110e+11	6.6166e+12	24719
## + HalfBath	1	5.4828e+11	6.6594e+12	24726
## + CentralAir	1	4.3689e+11	6.7708e+12	24744
## + BedroomAbvGr	1	2.5739e+11	6.9503e+12	24773
## + KitchenAbvGr	1	1.3210e+11	7.0756e+12	24792
## <none>			7.2077e+12	24806
## + OverallCond	1	2.7572e+10	7.1801e+12	24808
## + YrSold	1	1.4182e+10	7.1935e+12	24810
## + Id	1	1.0171e+10	7.1975e+12	24811

```

##
## Step: AIC=23710.75
## SalePrice ~ OverallQual
##
##          Df Sum of Sq      RSS   AIC
## + GrLivArea      1 7.7987e+11 1.8600e+12 23334
## + X1stFlrSF      1 6.1545e+11 2.0244e+12 23426
## + TotRmsAbvGrd   1 3.8077e+11 2.2591e+12 23547
## + GarageArea     1 3.6672e+11 2.2731e+12 23554
## + LotArea        1 3.0237e+11 2.3375e+12 23584
## + GarageCars     1 2.9868e+11 2.3412e+12 23586
## + Fireplaces     1 1.9849e+11 2.4414e+12 23632
## + FullBath       1 1.5636e+11 2.4835e+12 23651
## + YearRemodAdd   1 5.9554e+10 2.5803e+12 23693
## + X2ndFlrSF      1 5.6070e+10 2.5838e+12 23694
## + YearBuilt      1 5.1999e+10 2.5879e+12 23696
## + BedroomAbvGr   1 5.0790e+10 2.5891e+12 23696
## + HalfBath       1 2.8967e+10 2.6109e+12 23706
## + <none>                2.6399e+12 23711
## + CentralAir     1 4.9566e+09 2.6349e+12 23716
## + YrSold         1 2.7451e+09 2.6371e+12 23717
## + Id            1 2.0545e+09 2.6378e+12 23717
## + KitchenAbvGr   1 3.0071e+08 2.6396e+12 23718
## + OverallCond    1 1.6954e+08 2.6397e+12 23718
## - OverallQual    1 4.5678e+12 7.2077e+12 24806
##
## Step: AIC=23333.63
## SalePrice ~ OverallQual + GrLivArea
##
##          Df Sum of Sq      RSS   AIC
## + X1stFlrSF      1 2.5018e+11 1.6098e+12 23182
## + X2ndFlrSF      1 2.2963e+11 1.6304e+12 23196
## + GarageArea     1 2.0246e+11 1.6575e+12 23214
## + YearBuilt      1 1.8228e+11 1.6777e+12 23228
## + GarageCars     1 1.4881e+11 1.7112e+12 23249
## + BedroomAbvGr   1 1.3008e+11 1.7299e+12 23261
## + LotArea        1 1.1039e+11 1.7496e+12 23274
## + YearRemodAdd   1 8.9774e+10 1.7702e+12 23286
## + KitchenAbvGr   1 3.9493e+10 1.8205e+12 23317
## + Fireplaces     1 3.3799e+10 1.8262e+12 23320
## + CentralAir     1 2.5828e+10 1.8342e+12 23325
## + TotRmsAbvGrd   1 1.8578e+10 1.8414e+12 23330
## + HalfBath       1 1.8117e+10 1.8419e+12 23330
## + <none>                1.8600e+12 23334
## + Id            1 1.6321e+09 1.8584e+12 23340
## + YrSold         1 7.7611e+08 1.8592e+12 23340
## + FullBath       1 1.5251e+08 1.8598e+12 23340
## + OverallCond    1 1.0533e+08 1.8599e+12 23341
## - GrLivArea      1 7.7987e+11 2.6399e+12 23711
## - OverallQual    1 1.4137e+12 3.2737e+12 23947
##
## Step: AIC=23182.16
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF
##

```

```

##          Df Sum of Sq      RSS      AIC
## + YearBuilt      1 1.4114e+11 1.4687e+12 23088
## + GarageArea      1 1.1360e+11 1.4962e+12 23109
## + YearRemodAdd    1 8.9261e+10 1.5205e+12 23127
## + GarageCars      1 8.7310e+10 1.5225e+12 23128
## + BedroomAbvGr    1 8.4463e+10 1.5253e+12 23130
## + KitchenAbvGr    1 6.8749e+10 1.5411e+12 23141
## + LotArea         1 5.7740e+10 1.5521e+12 23149
## + CentralAir      1 1.6956e+10 1.5928e+12 23178
## + HalfBath        1 1.6947e+10 1.5929e+12 23178
## + TotRmsAbvGrd    1 1.4013e+10 1.5958e+12 23180
## + X2ndFlrSF       1 1.0678e+10 1.5991e+12 23182
## + Fireplaces      1 1.0413e+10 1.5994e+12 23182
## <none>                                1.6098e+12 23182
## + OverallCond     1 5.1535e+09 1.6046e+12 23186
## + Id              1 1.8068e+09 1.6080e+12 23188
## + YrSold          1 1.1149e+09 1.6087e+12 23188
## + FullBath        1 2.1842e+08 1.6096e+12 23189
## - X1stFlrSF       1 2.5018e+11 1.8600e+12 23334
## - GrLivArea       1 4.1460e+11 2.0244e+12 23426
## - OverallQual     1 1.1075e+12 2.7173e+12 23750
##
## Step:  AIC=23088.49
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt
##
##          Df Sum of Sq      RSS      AIC
## + BedroomAbvGr    1 8.1383e+10 1.3873e+12 23033
## + KitchenAbvGr    1 6.5353e+10 1.4033e+12 23046
## + LotArea         1 6.1704e+10 1.4070e+12 23048
## + OverallCond     1 5.7453e+10 1.4112e+12 23052
## + GarageArea      1 5.6093e+10 1.4126e+12 23053
## + GarageCars      1 3.0749e+10 1.4379e+12 23072
## + FullBath        1 2.8789e+10 1.4399e+12 23074
## + YearRemodAdd    1 2.5236e+10 1.4434e+12 23076
## + Fireplaces      1 1.7214e+10 1.4514e+12 23083
## + TotRmsAbvGrd    1 1.0446e+10 1.4582e+12 23088
## <none>                                1.4687e+12 23088
## + X2ndFlrSF       1 1.5606e+09 1.4671e+12 23094
## + Id              1 1.2813e+09 1.4674e+12 23094
## + HalfBath        1 7.6763e+08 1.4679e+12 23095
## + CentralAir      1 6.5649e+08 1.4680e+12 23095
## + YrSold          1 5.9223e+08 1.4681e+12 23095
## - YearBuilt      1 1.4114e+11 1.6098e+12 23182
## - X1stFlrSF      1 2.0904e+11 1.6777e+12 23228
## - OverallQual    1 4.5892e+11 1.9276e+12 23380
## - GrLivArea      1 5.1073e+11 1.9794e+12 23409
##
## Step:  AIC=23032.96
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +
##   BedroomAbvGr
##
##          Df Sum of Sq      RSS      AIC
## + LotArea         1 6.2522e+10 1.3248e+12 22989
## + OverallCond     1 6.1741e+10 1.3255e+12 22990

```

```

## + KitchenAbvGr 1 5.6535e+10 1.3307e+12 22994
## + GarageArea 1 4.5453e+10 1.3418e+12 23003
## + GarageCars 1 2.3081e+10 1.3642e+12 23022
## + FullBath 1 1.6326e+10 1.3709e+12 23027
## + YearRemodAdd 1 1.5700e+10 1.3716e+12 23028
## + Fireplaces 1 1.0635e+10 1.3766e+12 23032
## <none> 1.3873e+12 23033
## + X2ndFlrSF 1 3.0503e+09 1.3842e+12 23038
## + CentralAir 1 1.7023e+09 1.3856e+12 23039
## + TotRmsAbvGrd 1 1.0514e+09 1.3862e+12 23039
## + YrSold 1 7.3566e+08 1.3865e+12 23039
## + Id 1 5.6152e+08 1.3867e+12 23040
## + HalfBath 1 4.2095e+07 1.3872e+12 23040
## - BedroomAbvGr 1 8.1383e+10 1.4687e+12 23088
## - YearBuilt 1 1.3806e+11 1.5253e+12 23130
## - X1stFlrSF 1 1.6923e+11 1.5565e+12 23152
## - OverallQual 1 3.6408e+11 1.7514e+12 23282
## - GrLivArea 1 5.5910e+11 1.9464e+12 23397
##
## Step: AIC=22989.37
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +
## BedroomAbvGr + LotArea
##
## Df Sum of Sq RSS AIC
## + OverallCond 1 5.4224e+10 1.2705e+12 22950
## + KitchenAbvGr 1 4.6223e+10 1.2785e+12 22957
## + GarageArea 1 3.7101e+10 1.2877e+12 22965
## + GarageCars 1 1.7448e+10 1.3073e+12 22982
## + YearRemodAdd 1 1.5016e+10 1.3097e+12 22984
## + FullBath 1 1.4826e+10 1.3099e+12 22984
## <none> 1.3248e+12 22989
## + Fireplaces 1 4.0513e+09 1.3207e+12 22993
## + X2ndFlrSF 1 2.2243e+09 1.3225e+12 22994
## + TotRmsAbvGrd 1 2.1955e+09 1.3226e+12 22994
## + CentralAir 1 6.6297e+08 1.3241e+12 22996
## + YrSold 1 3.9053e+08 1.3244e+12 22996
## + Id 1 2.0118e+08 1.3246e+12 22996
## + HalfBath 1 6.5241e+07 1.3247e+12 22996
## - LotArea 1 6.2522e+10 1.3873e+12 23033
## - BedroomAbvGr 1 8.2201e+10 1.4070e+12 23048
## - X1stFlrSF 1 1.2633e+11 1.4511e+12 23082
## - YearBuilt 1 1.4200e+11 1.4667e+12 23094
## - OverallQual 1 3.8668e+11 1.7114e+12 23263
## - GrLivArea 1 4.9887e+11 1.8236e+12 23333
##
## Step: AIC=22950.52
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +
## BedroomAbvGr + LotArea + OverallCond
##
## Df Sum of Sq RSS AIC
## + KitchenAbvGr 1 3.5189e+10 1.2353e+12 22927
## + GarageArea 1 3.5147e+10 1.2354e+12 22927
## + GarageCars 1 1.7736e+10 1.2528e+12 22942
## + FullBath 1 1.3148e+10 1.2574e+12 22946

```

```

## <none> 1.2705e+12 22950
## + Fireplaces 1 3.6154e+09 1.2669e+12 22954
## + TotRmsAbvGrd 1 3.2341e+09 1.2673e+12 22955
## + CentralAir 1 2.2357e+09 1.2683e+12 22956
## + X2ndFlrSF 1 1.4131e+09 1.2691e+12 22956
## + YearRemodAdd 1 1.3546e+09 1.2692e+12 22956
## + YrSold 1 1.0520e+09 1.2695e+12 22957
## + Id 1 3.3072e+08 1.2702e+12 22957
## + HalfBath 1 5.2399e+06 1.2705e+12 22958
## - OverallCond 1 5.4224e+10 1.3248e+12 22989
## - LotArea 1 5.5005e+10 1.3255e+12 22990
## - BedroomAbvGr 1 8.6190e+10 1.3567e+12 23016
## - X1stFlrSF 1 1.4095e+11 1.4115e+12 23059
## - YearBuilt 1 1.9195e+11 1.4625e+12 23098
## - OverallQual 1 3.0762e+11 1.5781e+12 23181
## - GrLivArea 1 5.2521e+11 1.7957e+12 23323
##
## Step: AIC=22926.71
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +
## BedroomAbvGr + LotArea + OverallCond + KitchenAbvGr
##
## Df Sum of Sq RSS AIC
## + GarageArea 1 3.7118e+10 1.1982e+12 22900
## + GarageCars 1 2.1811e+10 1.2135e+12 22914
## + TotRmsAbvGrd 1 1.2850e+10 1.2225e+12 22922
## <none> 1.2353e+12 22927
## + CentralAir 1 6.2453e+09 1.2291e+12 22928
## + FullBath 1 5.7701e+09 1.2296e+12 22929
## + X2ndFlrSF 1 2.6668e+09 1.2327e+12 22931
## + Fireplaces 1 1.2374e+09 1.2341e+12 22933
## + YearRemodAdd 1 1.1883e+09 1.2342e+12 22933
## + YrSold 1 6.9978e+08 1.2346e+12 22933
## + Id 1 4.5523e+08 1.2349e+12 22933
## + HalfBath 1 1.3264e+08 1.2352e+12 22934
## - KitchenAbvGr 1 3.5189e+10 1.2705e+12 22950
## - OverallCond 1 4.3189e+10 1.2785e+12 22957
## - LotArea 1 4.7178e+10 1.2825e+12 22961
## - BedroomAbvGr 1 7.8317e+10 1.3137e+12 22987
## - X1stFlrSF 1 1.5813e+11 1.3935e+12 23052
## - YearBuilt 1 1.8035e+11 1.4157e+12 23069
## - OverallQual 1 2.5296e+11 1.4883e+12 23124
## - GrLivArea 1 5.4864e+11 1.7840e+12 23323
##
## Step: AIC=22900.25
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +
## BedroomAbvGr + LotArea + OverallCond + KitchenAbvGr + GarageArea
##
## Df Sum of Sq RSS AIC
## + TotRmsAbvGrd 1 1.0361e+10 1.1879e+12 22898
## <none> 1.1982e+12 22900
## + CentralAir 1 7.3618e+09 1.1909e+12 22900
## + FullBath 1 5.8701e+09 1.1924e+12 22902
## + X2ndFlrSF 1 2.9525e+09 1.1953e+12 22904
## + Fireplaces 1 1.8239e+09 1.1964e+12 22906

```

```
## + YearRemodAdd 1 1.4747e+09 1.1967e+12 22906
## + Id 1 8.1008e+08 1.1974e+12 22906
## + YrSold 1 5.2751e+08 1.1977e+12 22907
## + GarageCars 1 8.2145e+07 1.1981e+12 22907
## + HalfBath 1 2.8317e+07 1.1982e+12 22907
## - GarageArea 1 3.7118e+10 1.2353e+12 22927
## - KitchenAbvGr 1 3.7159e+10 1.2354e+12 22927
## - LotArea 1 3.9966e+10 1.2382e+12 22929
## - OverallCond 1 4.1145e+10 1.2394e+12 22930
## - BedroomAbvGr 1 6.8461e+10 1.2667e+12 22954
## - YearBuilt 1 1.2425e+11 1.3225e+12 23002
## - X1stFlrSF 1 1.2515e+11 1.3234e+12 23002
## - OverallQual 1 2.1877e+11 1.4170e+12 23077
## - GrLivArea 1 4.8757e+11 1.6858e+12 23268
##
## Step: AIC=22897.72
## SalePrice ~ OverallQual + GrLivArea + X1stFlrSF + YearBuilt +
## BedroomAbvGr + LotArea + OverallCond + KitchenAbvGr + GarageArea +
## TotRmsAbvGrd
##
## Df Sum of Sq RSS AIC
## <none> 1.1879e+12 22898
## + CentralAir 1 7.3575e+09 1.1805e+12 22898
## + FullBath 1 5.5856e+09 1.1823e+12 22900
## - TotRmsAbvGrd 1 1.0361e+10 1.1982e+12 22900
## + X2ndFlrSF 1 3.3058e+09 1.1846e+12 22902
## + Fireplaces 1 1.8307e+09 1.1860e+12 22903
## + YearRemodAdd 1 1.0114e+09 1.1868e+12 22904
## + Id 1 9.4975e+08 1.1869e+12 22904
## + YrSold 1 5.3972e+08 1.1873e+12 22904
## + GarageCars 1 1.7513e+08 1.1877e+12 22905
## + HalfBath 1 5.1538e+07 1.1878e+12 22905
## - GarageArea 1 3.4628e+10 1.2225e+12 22922
## - LotArea 1 4.1413e+10 1.2293e+12 22928
## - OverallCond 1 4.1555e+10 1.2294e+12 22928
## - KitchenAbvGr 1 4.5535e+10 1.2334e+12 22932
## - BedroomAbvGr 1 7.7845e+10 1.2657e+12 22960
## - X1stFlrSF 1 1.2544e+11 1.3133e+12 23001
## - YearBuilt 1 1.2784e+11 1.3157e+12 23003
## - OverallQual 1 2.1117e+11 1.3990e+12 23070
## - GrLivArea 1 2.4514e+11 1.4330e+12 23096
```

```
# The best stepwise regression model (10 predictors):
best.model1$call
```

```
## lm(formula = SalePrice ~ OverallQual + GrLivArea + X1stFlrSF +
## YearBuilt + BedroomAbvGr + LotArea + OverallCond + KitchenAbvGr +
## GarageArea + TotRmsAbvGrd, data = train.data)
```

```
# Coefficients of fitted model:
best.model1$coefficients
```

```
## (Intercept) OverallQual GrLivArea X1stFlrSF YearBuilt
```

```
## -1.069990e+06 1.664525e+04 6.456689e+01 3.694842e+01 5.073670e+02
## BedroomAbvGr LotArea OverallCond KitchenAbvGr GarageArea
## -1.464607e+04 7.430582e-01 6.103115e+03 -3.214772e+04 3.523431e+01
## TotRmsAbvGrd
## 4.054139e+03
```

```
# prediction:
pred1 = predict(best.model1, newdata = test.data)
# RMSE:
# Assuming 'actuals' is the actual values of the response variable in your testing dataset
rmse1 = sqrt(mean((pred1 - test.data$SalePrice)^2))
print(paste("RMSE:", rmse1))
```

```
## [1] "RMSE: 48820.768871431"
```

```
# R^2:
r_squared1 = cor(test.data$SalePrice, pred1)^2
print(paste("R^2:", r_squared1))
```

```
## [1] "R^2: 0.659587940625721"
```

```
# 3) Best subset BIC, 30 pts
best.model2 = regsubsets(SalePrice ~ ., data=train.data, nvmax=ncol(train.data)-1, method="exhaustive")
best.model2.summary = summary(best.model2)
best.size = which.min(best.model2.summary$bic)
# Coefficients of fitted model:
coef(best.model2, id=best.size)
```

```
## (Intercept) LotArea OverallQual OverallCond YearBuilt
## -1.017464e+06 7.339645e-01 1.667461e+04 5.943478e+03 4.812917e+02
## X1stFlrSF X2ndFlrSF BedroomAbvGr KitchenAbvGr TotRmsAbvGrd
## 1.013470e+02 6.477953e+01 -1.488137e+04 -3.342536e+04 4.361293e+03
## GarageArea
## 3.559750e+01
```

```
formula_best = as.formula(paste("SalePrice ~", paste(names(coef(best.model2, id=best.size))[-1], collapse=" + "))
# The best subset model based on BIC (all 10 predictors)
print(formula_best)
```

```
## SalePrice ~ LotArea + OverallQual + OverallCond + YearBuilt +
## X1stFlrSF + X2ndFlrSF + BedroomAbvGr + KitchenAbvGr + TotRmsAbvGrd +
## GarageArea
```

```
fitted_model = lm(formula_best, data=train.data)
# prediction:
pred2 = predict(fitted_model, test.data)
# RMSE:
# Assuming 'actuals' is the actual values of the response variable in your testing dataset
rmse2 = sqrt(mean((pred2 - test.data$SalePrice)^2))
print(paste("RMSE:", rmse2))
```

```
## [1] "RMSE: 48727.6803068858"
```

```
# R^2:  
r_squared2 = cor(test.data$SalePrice, pred2)^2  
print(paste("R^2:", r_squared2))
```

```
## [1] "R^2: 0.660115195606302"
```

```
# 4) 25 pts  
# Compare stepwise BIC and bestsubset SSE:  
rmse1
```

```
## [1] 48820.77
```

```
rmse2# win
```

```
## [1] 48727.68
```

```
r_squared1
```

```
## [1] 0.6595879
```

```
r_squared2# win
```

```
## [1] 0.6601152
```

```
(bic1 = BIC(best.model1))
```

```
## [1] 26017.87
```

```
(bic2 = BIC(fitted_model))# win
```

```
## [1] 26016.31
```

```
# Stepwise based on BIC in part 3) is better! according to RMSE, R square, and BIC
```

There are several things we can do to further improve our model: 1. Outlier removal Outliers can significantly affect the performance of a linear regression model. Use diagnostic plots (e.g., Cook's distance) to identify and investigate outliers. Removing or appropriately adjusting outliers can improve model accuracy. 2. Transformation Non-linear relationships between predictors and the response variable might not be well captured by a linear model. 3. Multicollinearity High correlation among predictors can inflate variance estimates, making coefficients unreliable.